

MAHADI HASSAN

Technical Lead | Site Reliability Engineer

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PROFESSIONAL SUMMARY

Technical Lead with 5+ years of experience building and supporting production systems across AWS-hosted backend services, IoT/cloud platforms, and customer-facing web/mobile applications. Hands-on engineer with strong experience in incident triage, debugging across application, data, and infrastructure layers, guiding Level 2 support operations, and improving reliability through automation and disciplined engineering. Additional hands-on exposure includes LLM tooling and workflow automation with OpenAI and Claude, along with vector database-backed retrieval patterns. Strong background in Node.js, NestJS, Go, Python, gRPC, GraphQL, SQL, Redis, Docker, Terraform, Ansible, Linux, and CI/CD.

CORE SKILLS

- **Production Support & Reliability:** incident triage, root cause investigation, customer-impacting issue resolution, operational documentation, support escalation
- **Backend & Services:** Node.js, NestJS, Go, Python, gRPC, GraphQL, REST APIs, microservices
- **AI & Automation:** OpenAI, Claude, prompt workflows, automation tooling, vector databases, retrieval-augmented patterns
- **Cloud & Infrastructure:** AWS, Terraform, Ansible, Docker, Linux, CI/CD
- **Data & Messaging:** MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, RabbitMQ, MQTT
- **Frontend & Client Debugging:** JavaScript, React, Next.js, Angular, React Native, Ionic, Expo
- **Systems:** IoT platforms, edge/cloud integration, Modbus/TCP, LoRaWAN

PROFESSIONAL EXPERIENCE

SAVE ENERGY SYSTEMS, INC.

Associate Technical Lead

Dec 2024 - Present

- Guide hands-on delivery across backend, mobile, and Level 2 support operations for production cloud and edge systems, helping the team triage and resolve complex customer-impacting issues.
- Mentor 5+ junior engineers on debugging approach, implementation quality, and issue ownership across active production work.
- Design scalable, fault-tolerant solutions for core energy management applications used by enterprise clients.
- Improve resilience in Demand Limiting Controller workflows by contributing to complex edge fault handling for HVAC control systems.
- Implement Over-the-Air updates over VPN for remote edge devices, reducing manual field intervention and improving operational speed.
- Automate DLC replacement and edge provisioning with Ansible, reducing deployment time by approximately 40% while lowering manual errors.
- Create AI-assisted operational workflows for incident handling, alerting, escalation, and Prometheus/Grafana dashboard support.
- Integrate Modbus/TCP, MQTT, and LoRaWAN communication flows to support real-time device connectivity and control.

Senior Software Engineer

Jun 2024 - Dec 2024

- Contributed to building a scalable IoT platform managing 6,000+ edge devices across 900+ sites in the United States, enabling real-time energy optimization.
- Developed and maintained 10+ microservices for device communication, data ingestion, and control logic, processing thousands of MQTT events daily.
- Worked in a 10+ year-old legacy edge codebase and integrated legacy systems with a centralized cloud platform for monitoring and control.
- Provisioned AWS infrastructure with Terraform to support repeatable deployments and reliable cloud resource management.
- Automated infrastructure and deployment workflows using Ansible and CI/CD pipelines.

- Diagnosed issues across device, network, and cloud layers while maintaining compatibility with older Ubuntu-based edge environments.

SURGE GLOBAL

Senior Software Engineer (Full Stack/Mobile)

Sep 2023 - Jun 2024

- Revamped core web and mobile features, improving load performance by 30%+ and supporting application scalability for growing user traffic.
- Contributed to developing internal Node.js libraries to accelerate feature development and improve code reuse across products.
- Diagnosed and resolved critical issues across web and mobile platforms, supporting stable releases and high availability.
- Established CI/CD pipelines and AWS infrastructure as code with Terraform to improve deployment consistency and engineering velocity.

Software Engineer (Full Stack/Mobile)

Aug 2022 - Aug 2023

- Contributed to building and shipping core features across web and mobile applications used by thousands of users.
- Designed scalable solutions across frontend, backend, and infrastructure layers, reducing recurring support issues.
- Developed reusable private Node.js libraries to reduce duplication and speed up product delivery.
- Diagnosed and resolved critical bugs across application layers to support stable release cycles.
- Created CI/CD workflows and infrastructure as code with Terraform.

INGENII IOT

Intern Software Engineer / Associate Software Engineer

Jan 2021 - Aug 2022

- Contributed to building reporting, task management, user management, and dynamic data visualization tools for web and Excel-based workflows.
- Developed RESTful services and integrated third-party APIs based on project requirements.
- Worked with IoT sensor data and MQTT-based systems to deliver end-to-end solutions.
- Improved application performance through SQL query optimization and caching strategies.
- Resolved UI bugs, API issues, and critical memory leaks in the web application.
- Collaborated directly with clients and stakeholders to gather requirements and solve technical issues.

EDUCATION

Sri Lanka Institute of Information Technology

BSc in Information Technology, specializing in Software Engineering | Feb 2018 - Dec 2022

- Second Class Upper Honours

Zahira College

GCE Advanced Level | 2018

- Subjects: Combined Mathematics, Physics, Chemistry

CERTIFICATIONS & AWARDS

- AWS Certified Developer - Associate (in progress)
- Runner-up, Codefest Hackathon 2019
- Finalist, Hack-Bit 2020

LANGUAGES

- English - Professional proficiency
- Sinhala - Native